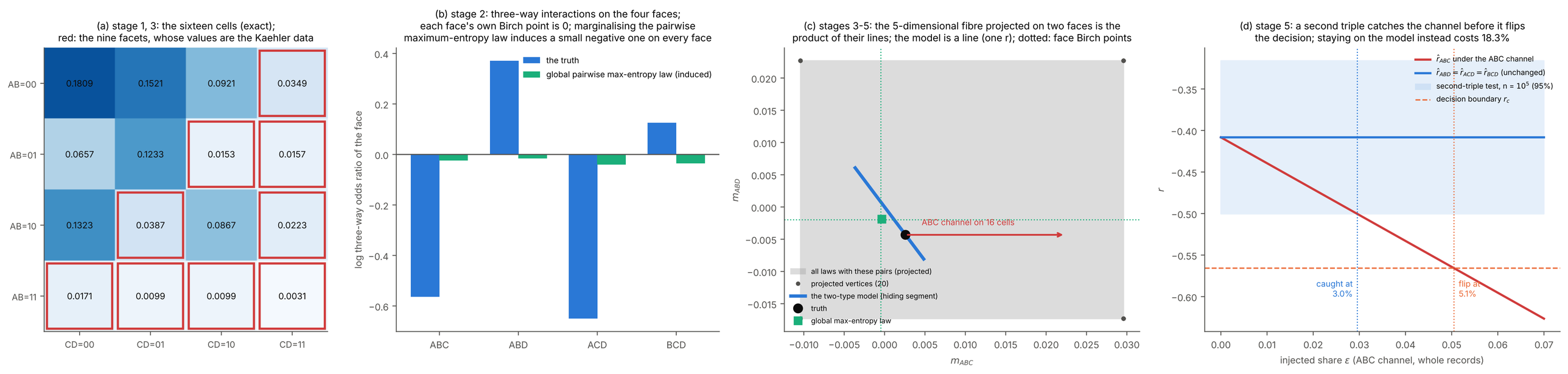
(a) stage 1, 3: the sixteen cells (exact);
red: the nine facets, whose values are the Kaehler data
AB=00
AB=01
AB=10
AB=11
0.1809
0.1521
0.0921
0.0349
0.0657
0.1233
0.0153
0.0157
0.1323
0.0387
0.0867
0.0223
0.0171
0.0099
0.0099
0.0031
CD=00
CD=01
CD=10
CD=11
(b) stage 2: three-way interactions on the four faces;
each face's own Birch point is 0; marginalising the pairwise
maximum-entropy law induces a small negative one on every face
the truth
global pairwise max-entropy law (induced)
log three-way odds ratio of the face
0.4
0.2
0.0
−0.2
−0.4
−0.6
ABC
ABD
ACD
BCD
(c) stages 3-5: the 5-dimensional fibre projected on two faces is the
product of their lines; the model is a line (one r); dotted: face Birch points
ABC channel on 16 cells
all laws with these pairs (projected)
projected vertices (20)
the two-type model (hiding segment)
truth
global max-entropy law
m_{ABD}
m_{ABC}
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015
−0.010
−0.005
0.000
0.005
0.010
0.015
0.020
0.025
0.030
(d) stage 5: a second triple catches the channel before it flips
the decision; staying on the model instead costs 18.3%
$\hat{r}_{ABC}$ under the ABC channel
$\hat{r}_{ABD} = \hat{r}_{ACD} = \hat{r}_{BCD}$ (unchanged)
second-triple test, n = 10^5 (95%)
decision boundary r_c
caught at
3.0%
flip at
5.1%
r
−0.35
−0.40
−0.45
−0.50
−0.55
−0.60
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
injected share ε (ABC channel, whole records)